Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: Shishe (Nikolay Bandura, Ashley Bandura)
b. Team leader name: Nikolay Bandura
c. Problem Statement: Retail & Commerce. Brands pay for in-store promotions they cannot verify; StoreOps AI checks, explains and fixes them with Gemini.

Brief about the idea

The problem

Brands pay retailers for promotions: shelf space, facings, price cards. Nobody can easily check that it happened in each store. Stock sits in the back room, sales drop, and the promotion budget is wasted.

StoreOps AI

- Gemini turns the vendor agreement into checkable rules, each with the quote and page it came from. A person approves them.
- From one shelf photo plus sales and stock data, the Investigator explains why the promotion is failing and cites its evidence.
- The rep gets concrete actions; an after-photo is checked against every approved rule. Only a full PASS closes the visit.

Live on Google Cloud: Gemini 3.8 Flash, Cloud Run, Firestore (Firebase), Cloud Storage, BigQuery.



Before photo from the demo: the agreement requires 6 facings, the shelf has 1.

Opportunities

How is it different from existing ideas?

Retail execution apps count products in photos; trade-promotion tools track contracts and spend. They rarely meet. StoreOps links the contract clause, the sales and stock data and the shelf photo in one evidence-backed case, and closes it only when a new photo proves the fix.

How will it solve the problem?

Rules come from the agreement (Gemini proposes, a person approves). When sales drop, the Investigator diagnoses the cause and proposes actions. The Verifier checks the after-photo rule by rule; PARTIAL or FAIL keeps the case open.

USP: verified, not claimed

- Every claim cites evidence; numbers are computed in code, never by the model.
- Model output is schema-validated and fails closed; only people approve rules.
- One phone photo is enough for the rep; brands get proof of execution store by store.

List of features offered by the solution

Agreement → rules

- PDF upload; Gemini extraction with quote and page
- Uncheckable clauses flagged as gaps, never guessed
- Human approval; locked, versioned policy

Sales & stock data

- CSV import, validation, SHA-256 fingerprint, commit
- Facts in BigQuery; trends computed in code

Roles

- Admin and Rep views; works on a phone

Investigation

- Gemini observes the shelf photo (facings per zone)
- Hypothesis with evidence-cited claims
- Concrete action plan for the rep

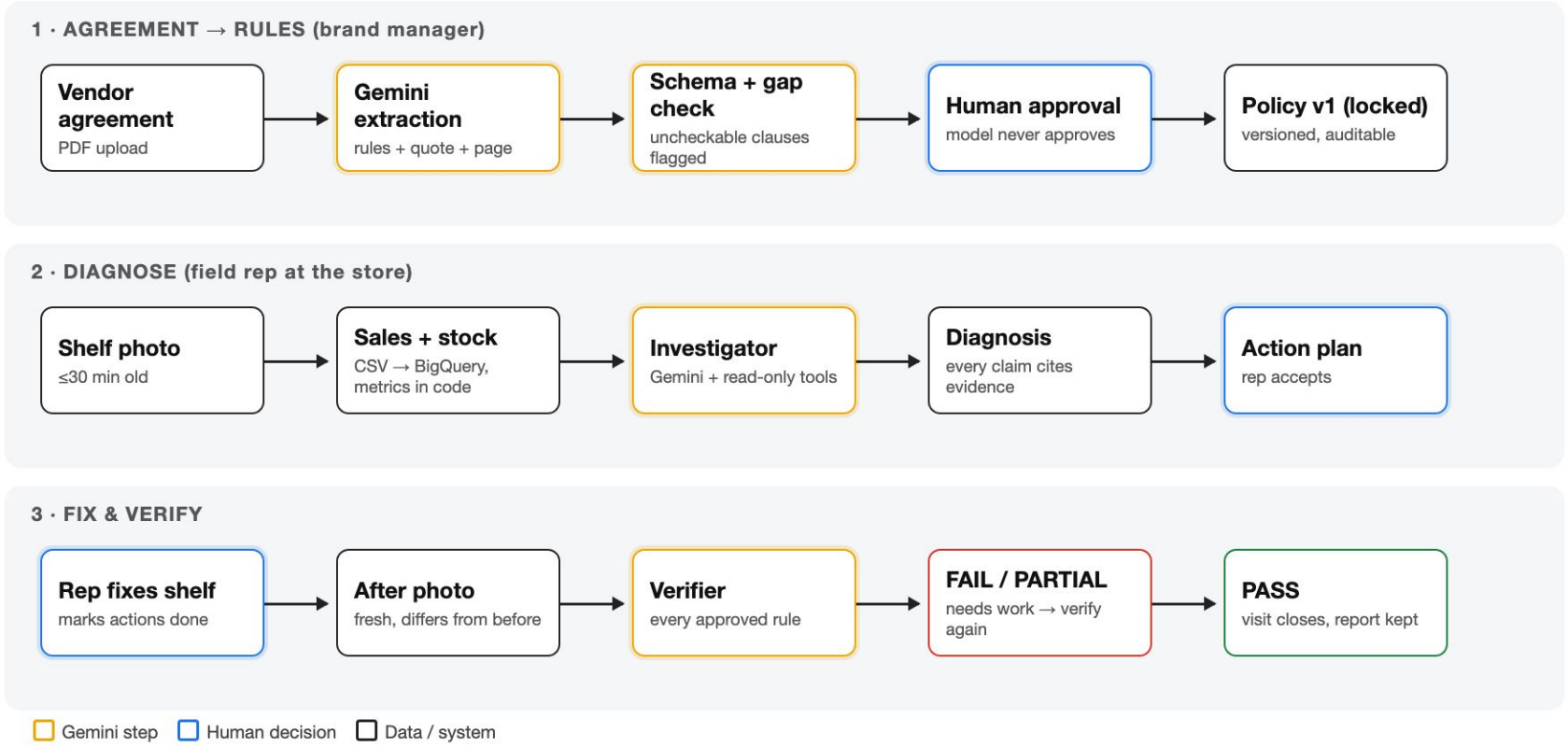
Fix & verify

- After-photo must be ≤30 min old and differ from before
- Rule-by-rule PASS / FAIL; verify again until PASS

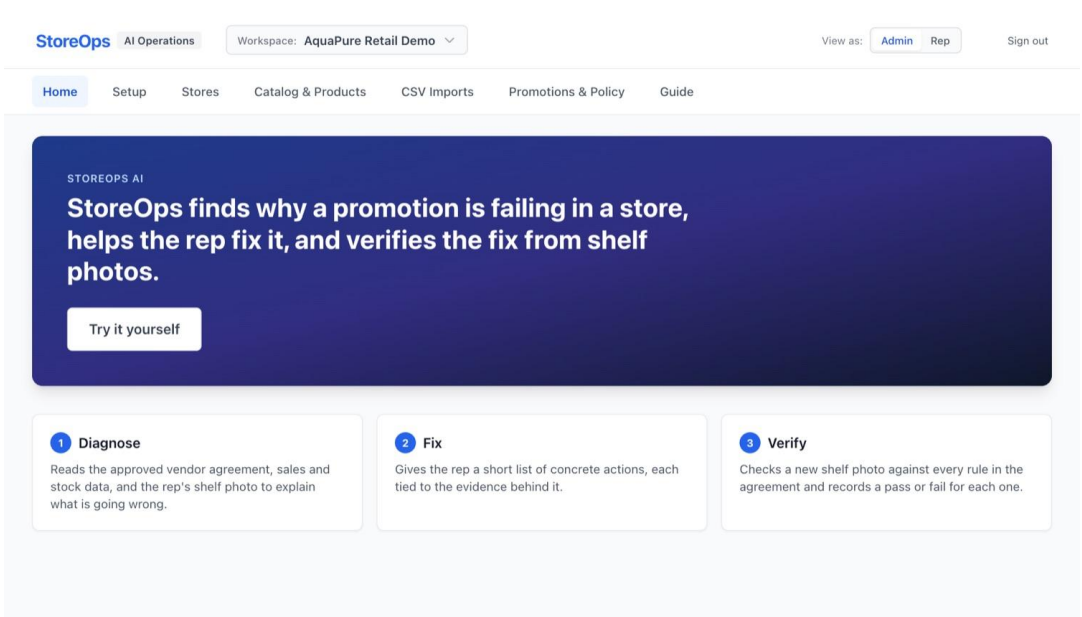
Trust & safety

- Schema validation, evidence grounding, prompt-injection-safe notes
- Stale-policy detection, signed expiring media links, deny-all Firestore client rules

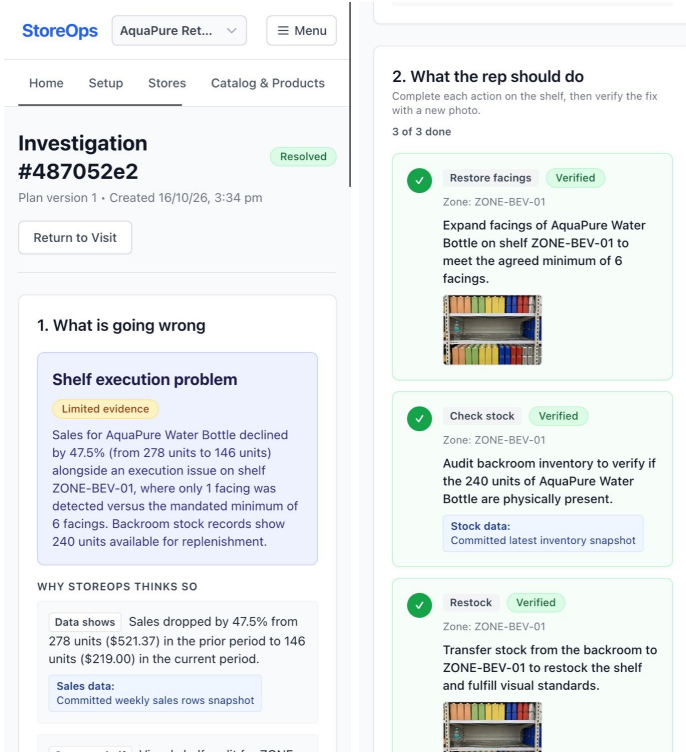
Process flow diagram



Wireframes / screens of the solution

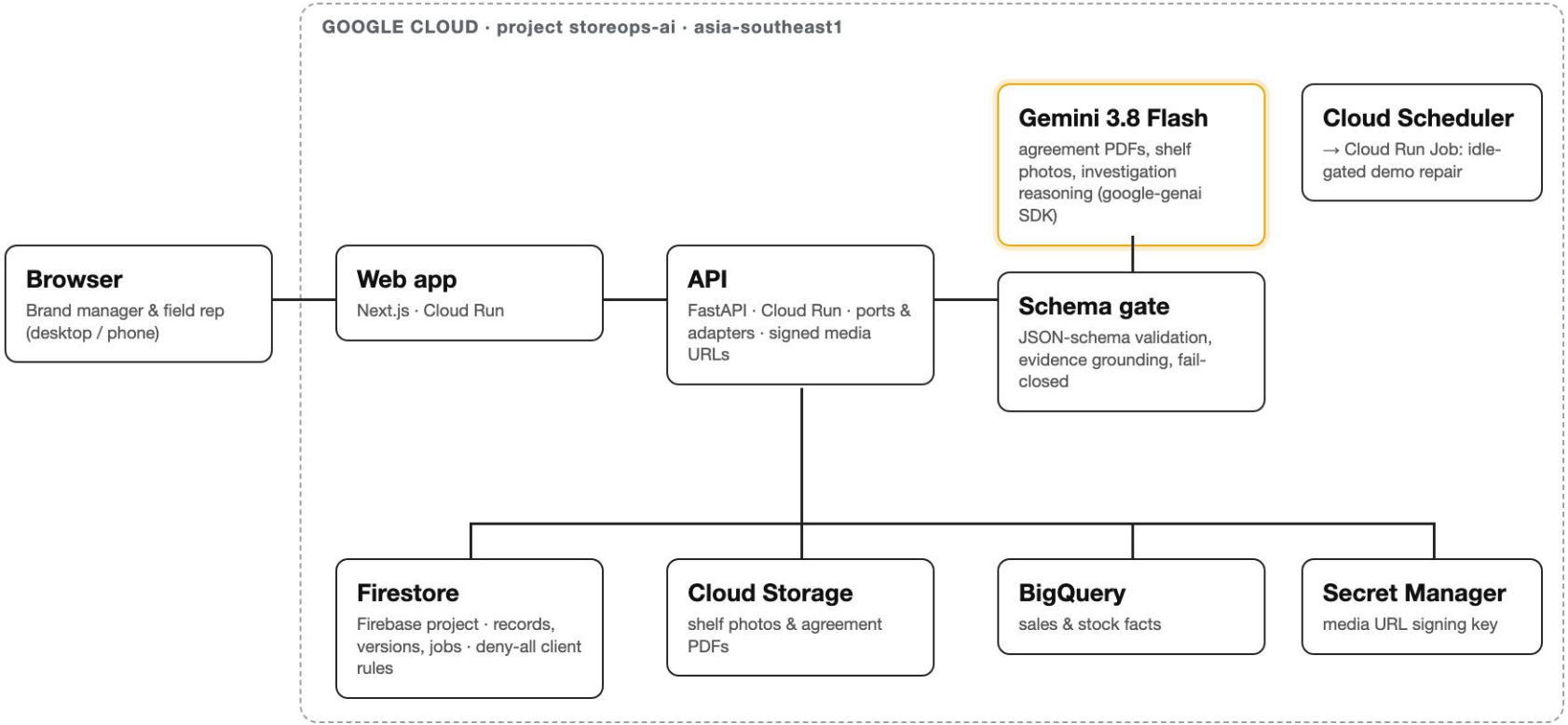


Landing page: one sentence, try it yourself



Rep on a phone: diagnosis and actions

Architecture diagram of the proposed solution



Technologies used in the solution

Generative AI

- Gemini 3.8 Flash (google-genai SDK): agreement PDF extraction, shelf-photo observation, investigation and verification reasoning
- Structured JSON output validated against shared schemas, one repair attempt, then fail closed

Google Cloud

- Cloud Run: web app, API, worker and the demo-repair job
- Firestore in our Firebase project: records, policy versions, jobs, idempotency (deny-all client rules)
- Cloud Storage: shelf photos and agreements · BigQuery: sales and stock facts
- Secret Manager, Cloud Scheduler, Cloud Build and Artifact Registry

Application

- Next.js + TypeScript web app; FastAPI + Python API; one OpenAPI contract shared by both
- 386 backend tests (pytest) and 59 web tests, all passing

Estimated implementation cost (optional)

What drives cost

- Gemini calls: per store visit, one investigation (a few model calls) and one or two verifications.
- Pay-per-use Google Cloud services: Cloud Run, Firestore, Cloud Storage and BigQuery scale with stores and visits.
- No per-seat software for reps: a phone browser is enough.

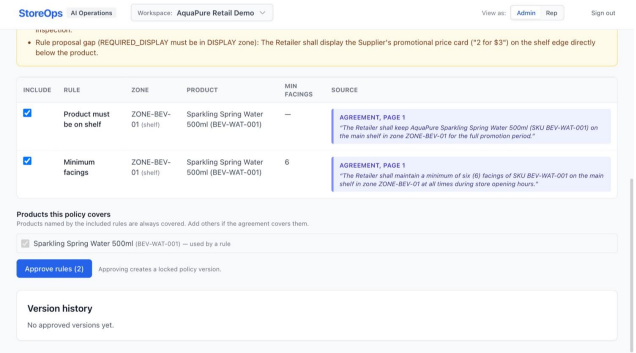
How we keep it low

- Low thinking level for shelf-photo observation: about 73 s → 6–8 s per photo, with fewer tokens.
- Numbers are computed in code and BigQuery, so the model only reasons over small, cited summaries.
- Background jobs scale to zero; the demo API runs as a single small Cloud Run instance.

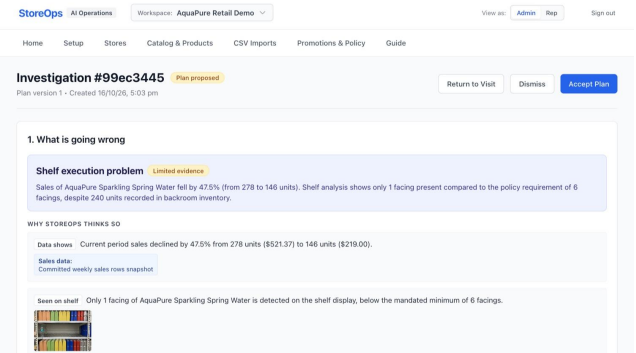
Next step

We have not yet measured cost per visit in production. A pilot with one brand and a few stores would give that figure.

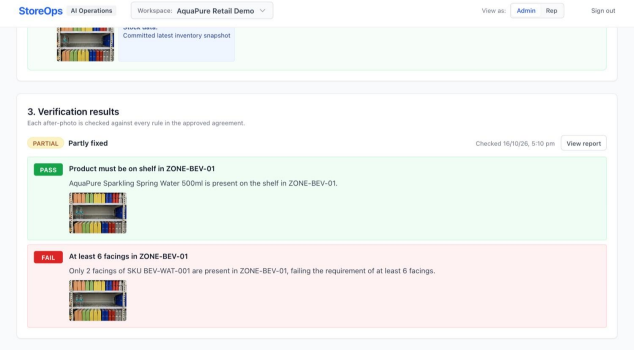
Snapshots of the prototype



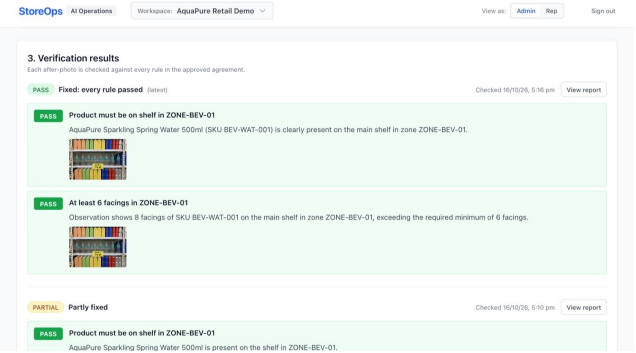
Gemini proposes rules with quote and page



Diagnosis: −47.5% sales, 1 of 6 facings, 240 in back room



After-photo 1: FAIL, only 2 facings



After-photo 2: PASS, 8 facings, visit closes

Prototype performance report / benchmarking

Measured on the live deployment, Oct 16 2026	Result	Notes
Rule extraction from the agreement PDF	**19 s**	2 rules + 3 flagged gaps
Investigation (photo + sales + stock)	**≈ 1 min**	EXECUTION hypothesis, 3 actions
Verification per after-photo	**16–28 s**	was ≈ 140 s before tuning
Shelf-photo observation call	**6–8 s**	was ≈ 73 s; thinking level LOW
Full example: diagnose → FAIL → PASS	**1 min 44 s**	visit resolved automatically
Automated tests	**386 + 59**	backend + web, all passing

Times include Gemini 3.8 Flash calls made from Cloud Run in asia-southeast1.

Additional details / future development

- Real sign-in with Firebase Authentication and one workspace per brand (the demo uses a shared token).
- Scale-out: read Firestore directly to remove today's single-instance limit.
- More rule types: price cards and displays (already detected as gaps today), planogram images.
- Retailer data connectors instead of CSV files; alerts when a promoted product's sales drop.
- Brand dashboard: compliance by store, with promotion funding tied to verified periods (clause 5 of the sample agreement).

Honest limits today

- Demo brand, agreement and sales data are fictional; all app screens are real runs on the live deployment.
- The demo presenter in the video is AI-generated (Gemini image + Veo).

Provide links to your:

GitHub public repository

https://github.com/exbald/storeops-ai

Demo video (3:14)

https://youtu.be/KR8Y7qN2x9w

Final product link

https://storeops-web-967137522464.asia-southeast1.run.app

Open “See a completed example”, or “Try it yourself” to run a live Gemini investigation.

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

